Bachelor of Information Technology in Computer Systems Security (BCSS)
IPB49804 Final Year Project 1
Final Year Project Proposal Form

Final Year Project's Proposal Form (fill up by student)		
1.	Student's Name	MOHAMMAD AZRI BIN AZIZ
2.	Contact Number	011-3994-4101
3.	Current CGPA	3.77
4.	Working Title	SentriZK: Secure Internal Messaging with Zero-Knowledge Authentication and AI Anomaly Detection
5.	Background/Introduction	Internal messaging systems are widely used in organizations for daily communication. Although many use end-to-end encryption (E2EE) and multi-factor authentication (MFA), research shows these still expose credentials and lack real-time detection of suspicious behavior. Studies in IEEE highlight that anomaly detection is often applied to networks or IoT systems, but not embedded into messaging platforms. Similarly, Zero-Knowledge Proof (ZKP) based authentication has been proposed in access control but is rarely used in internal chat systems. SentriZK aims to fill this gap by integrating ZKP authentication and AI-based anomaly detection into a secure internal messaging framework, improving confidentiality, integrity, and availability.
6.	Problem Statement	**Authentication Vulnerability in Internal Systems:** Many organizations use internal chat platforms like Slack or Microsoft Teams that rely on centralized servers to store hashed credentials. In real cases such as the 2023 Slack data breach, attackers exploited stolen session tokens, exposing internal communications. This shows that even with MFA, stored credentials remain a weak point. Using Zero-Knowledge Proof (ZKP) authentication can eliminate this issue by verifying users without storing or transmitting passwords. **Lack of Intelligent Threat Detection in Messaging:** While end-to-end encryption protects message content, it prevents servers from analysing chats for threats. For example, in 2025 attackers hijacked expired Discord invite links and used them to deliver multi-stage malware and steal credentials, highlighting how malicious activity can travel inside normal chat flows and go undetected by centralized analysis, motivating on-device AI anomaly detection to identify risky behaviour locally without breaking encryption.
7.	Objectives	**To study** existing authentication and anomaly detection methods in internal messaging systems and identify their security limitations.

		To develop a secure internal chat application integrating Zero-Knowledge Proof authentication and on-device AI anomaly detection. **To test** the system's effectiveness in preventing unauthorized access and detecting suspicious behaviour without compromising privacy.
8.	Project Scope / Coverage	**Target Users:** Small and Medium Enterprises (SMEs) using internal office communication systems. **System Functions:** Provides secure user authentication using Zero-Knowledge Proof, enables end-to-end encrypted messaging, and performs on-device AI anomaly detection to identify suspicious messages or behaviors. **Platform:** Android-based mobile application with a Node.js backend for ZKP verification and messaging, and TensorFlow Lite for on-device AI processing.
9.	Literature Review (Proposal only - 3 articles)	**Enhancing Authentication using Zero-Knowledge Proofs** Bhattacharya et al. (2024) explore the application of Zero-Knowledge Proofs (ZKPs) in enhancing digital privacy within authentication systems. The paper discusses the theoretical foundations of ZKPs, including zk-SNARKs and zk-STARKs, and their ability to verify user credentials without disclosing personal information. The authors compare ZKP-based authentication systems with traditional methods, highlighting superior privacy preservation and scalability. This study underscores the potential of ZKPs in securing internal messaging platforms by ensuring user authentication without compromising privacy **Anomaly Detection Using Machine Learning Techniques** Natha et al. (2022) provide a comprehensive review of machine learning methods for anomaly detection, covering applications, performance metrics, and classification strategies across various domains. The study highlights the effectiveness of techniques such as Naïve Bayes, Support Vector Machine (SVM), and Autoencoders in identifying unusual patterns in data. The authors also discuss the challenges in dataset selection and the need for real-world data to train robust models. This review serves as a foundational reference for implementing AI-based anomaly detection in secure messaging systems, offering insights into suitable algorithms and evaluation criteria. **References** Bhattacharya, S., Seth, D., Panyam, S., & Gangrade, P. (2024). Enhancing digital privacy: The application of zero-knowledge proofs in authentication systems. International Journal of Computer Trends and Technology, 72(4), 34–41. https://doi.org/10.14445/22312803/IJCTT-V72I4P104 Natha, S., Leghari, M., Rajput, M. A., Zia, S. S., & Shabir, J. (2022). A systematic review of anomaly detection using machine and deep learning techniques. Quaid-e-Awam University Research Journal of Engineering, Science & Technology, 20(1), 83–94. https://doi.org/10.52584/QRJ.2001.11

10.	Contribution to Cybersecurity Domain	**System Security & Application Security:** SentriZK enhances internal messaging systems by securing authentication (ZKP) and monitoring for anomalies (AI), protecting the system from unauthorized access and insider threats. **Mobile Security:** Implements on-device AI anomaly detection, ensuring threats are detected locally without compromising user data.
11.	Proposed Supervisor Signature	

For Office Use Only		
1.	Evaluator's Comment:	☐ Accept ☐ Reject
2.	Comments/ Remarks	
3.	Assigned supervisor	
4.	Checked & Verified by: ..... Name: Designation: Date:	

(Note: Maximum page proposal 3 pages)